US DISASTER DECLARATIONS DASHBOARD

Technical Documentation, Data Model, DAX, Validation and Deployment

1. Project overview

The project uses the supplied US disaster-declaration data to build a five-page interactive Power BI report covering summary indicators, geography, temporal patterns, incident/program comparisons and frequency-based scenario analysis.

The primary analysis table is the cleaned full-history file US_Disaster_PowerBI_Ready.csv, loaded as Disaster (approximately 64,092 records in the current model). The separate us_disasters_m5.csv filtered subset was kept separate and not appended, to avoid potential double-counting.

2. Source data and preparation

• Main source: us_disaster_declarations.csv (full-history dataset). Additional supplied file: us_disasters_m5.csv (filtered subset).

• Prepared input: US_Disaster_PowerBI_Ready.csv. Preparation included trimming categorical text, converting date fields and adding derived columns.

• Derived columns: DisasterDuration, TimeToDeclaration, DeclarationYear, DeclarationMonthNumber and DeclarationMonth.

• Key source fields include disaster_number, state, declaration_type, declaration_date, incident_type, incident dates, designated_area/FIPS and program flags such as PA, IA and HM.

• Missing or unusable source dates produce blank derived duration values.

3. Data model and relationships

The implemented model has a central Disaster table and three lookup dimensions:

3.1 Relationship design

One-to-many relationships: DimState[state] → Disaster[state]; DimIncidentType[incident_type] → Disaster[incident_type]; DimDeclarationType[declaration_type] → Disaster[declaration_type]. Use single-direction filtering from dimension to Disaster. Do not use declaration_date as a one-side key because dates repeat.

3.2 ProgramTypes modelling gap

The assignment brief also refers to a ProgramTypes dimension. In the current implementation, assistance-program indicators remain separate binary columns in Disaster (for example, pa_program_declared, ia_program_declared and hm_program_declared). A normalized ProgramTypes dimension has not been created. This is a documented deviation. A future enhancement is to unpivot the program flag columns into a bridge/fact table with one row per disaster-program combination and relate it to DimProgramType.

Conceptual ERD: DimState (1) — (*) Disaster; DimIncidentType (1) — (*) Disaster; DimDeclarationType (1) — (*) Disaster. The frequency What-If parameter and any ScenarioAxis/FundingAssumption tables are disconnected helper tables, not part of the disaster star schema.

4. Power Query preparation

• Load the prepared full-history CSV into Power BI and name the query/table Disaster.

• Confirm date fields are Date data type and category fields are Text.

• Trim category text to prevent duplicate-looking values caused by leading/trailing spaces.

• Create dimension queries by referencing Disaster, retaining the relevant category column, removing duplicates and naming the queries DimState, DimIncidentType and DimDeclarationType.

• Keep the M5 filtered dataset separate; do not append it to Disaster.

• Check nulls in incident_begin_date, incident_end_date and declaration_date before interpreting duration calculations.

5. Derived columns and DAX measures

5.1 Derived columns

• DisasterDuration: days between incident_end_date and incident_begin_date.

• TimeToDeclaration: days between declaration_date and incident_begin_date.

• DeclarationYear: year extracted from declaration_date.

• DeclarationMonth: month label derived from declaration_date.

• DeclarationMonthNumber: numeric month used to sort DeclarationMonth chronologically.

These columns were prepared in the cleaned CSV. If recreated in Power Query or DAX, preserve blank results when required dates are missing.

5.2 Core DAX measures

```
Total Disasters: Total Disasters = COUNTROWS(Disaster)

Average Duration (Days): Average Duration (Days) = AVERAGE(Disaster[DisasterDuration])

Total PA Declarations: Total PA Declarations = CALCULATE(COUNTROWS(Disaster), Disaster[pa_program_declared] = 1)

Total HM Declarations: Total HM Declarations = CALCULATE(COUNTROWS(Disaster), Disaster[hm_program_declared] = 1)

Total IA Declarations: Total IA Declarations = CALCULATE(COUNTROWS(Disaster), Disaster[ia_program_declared] = 1)
```

5.3 Scenario DAX measures

```
Scenario Adjustment %: Scenario Adjustment % = DIVIDE(SELECTEDVALUE(frequency[Frequency Adjustment], 0), 100)

Projected Disasters: Projected Disasters = ROUND([Total Disasters] * (1 + [Scenario Adjustment %]), 0)

Projected IA Declarations: Projected IA Declarations = ROUND([Total IA Declarations] * (1 + [Scenario Adjustment %]), 0)

Projected PA Declarations: Projected PA Declarations = ROUND([Total PA Declarations] * (1 + [Scenario Adjustment %]), 0)
```

The frequency What-If parameter was configured from -20 to +20, increment 5, default 0. Replace the generated column reference in the formula if its actual name differs.

5.4 PA funding estimate and assumption

The supplied source data does not include actual PA funding dollars or expenditure. Therefore actual FEMA PA spending cannot be calculated from this dataset. If projected PA funding is displayed, define a clearly labelled input Average PA Funding per Declaration, state the currency and assumed value, and calculate: Projected PA Funding = [Projected PA Declarations] × [Average PA Funding per Declaration]. The assumed average must be supplied/approved or clearly marked as an illustrative demonstration value. Do not label it actual funding or an official forecast.

6. Report page inventory

7. Testing and validation

Complete the tests below and record actual results. The status is intentionally left as To complete; do not mark a test passed until it has been performed.

8. Publishing and refresh procedure

• Open the final PBIX in Power BI Desktop and save the latest version.

• Select Home → Publish, sign in with the relevant Power BI account and choose the approved workspace.

• Open Power BI Service and verify the report and semantic model are present and visuals load.

• If the CSV is stored locally, configure an on-premises data gateway for scheduled refresh, or move the source to an approved cloud location supported by the organisation.

• In semantic model settings, configure data source credentials and refresh schedule if available. Run a test refresh and confirm success.

• If no gateway, cloud source or refresh schedule is configured, describe the report as manually refreshable in Desktop; do not claim scheduled refresh is active.

9. Known limitations

• COUNTROWS counts loaded declaration records, not necessarily unique real-world disaster events.

• The M5 subset is excluded from the main report to avoid potential duplication.

• Program indicators remain separate flag columns; the dedicated ProgramTypes dimension is a future modelling enhancement.

• Actual PA funding amounts are absent; any funding visual is based on an explicit user-defined assumption.

• Scenario outputs are proportional sensitivity-analysis results, not predictive models or official forecasts.

• Missing dates produce blank duration/time-to-declaration values and may reduce the records included in averages and scatter plots.

10. Submission checklist

• Final PBIX saved with a clear filename.

• One-page User Guide included.

• Technical Documentation included.

• 50-record validation completed and results recorded.

• Power BI Service publishing completed, if required.

• PA funding assumption and modelling limitations disclosed.

Table	Key / attribute	Role
Disaster	disaster_number and descriptive/date/program fields	Central record table used by report measures and visuals.
DimState	state (unique)	State lookup.
DimIncidentType	incident_type (unique)	Incident category lookup.
DimDeclarationType	declaration_type (unique)	Declaration category lookup.

Page	Visuals / purpose
1. Overview	Five KPI cards: Total Disasters, Average Duration, Total PA, Total HM and Total IA; date/year and incident-type slicers.
2. Geographic Analysis	US state map and Top 10 designated areas/counties bar chart.
3. Temporal Trends	Monthly disaster trend line chart and program declaration area chart.
4. Incident & Program Insights	Clustered bar comparing program declarations by incident type; scatter plot of DisasterDuration vs TimeToDeclaration.
5. Scenario Insights	Frequency What-If slicer, projected disaster/IA outputs and PA funding estimate subject to assumption.

Test	Procedure / expected result	Status
Row count	Confirm Disaster contains 64,092 rows (count observed in current model).	To complete
50-record sample	Select 50 random records and compare disaster_number, state, declaration_date, incident_type and program flags with source CSV.	To complete
Date types	Confirm declaration_date, incident_begin_date and incident_end_date are Date fields, not text.	To complete
Duration	For sampled rows with both incident dates, verify end date minus begin date in days; missing dates remain blank.	To complete
Time to declaration	Verify declaration_date minus incident_begin_date in days for sampled rows with both dates.	To complete
Program flags	Compare sampled PA/IA/HM flags with source; check measures respond to filters.	To complete
Relationships	Confirm unique dimension keys and one-to-many, single-direction relationships.	To complete
Slicers and visuals	Change date and incident-type slicers; verify visuals update without errors.	To complete
Scenario	Test -20%, 0% and +20%; outputs change proportionately and return to baseline at 0%.	To complete
Visual review	Check errors, clipped titles, readable labels and chronological month sorting.	To complete